

Hysterical Conformal Cyclic Cosmology

Infrared electromagnetic criticality and an active big-bang crossover

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Working mathematical formulation — September 2026

Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

The late-universe gravitational response of this series forces dilution-driven criticality and a vacuum-like attractor. Here we isolate a companion late-time channel: infrared electromagnetic response in an asymptotically de Sitter background. A minimal helical three-mode linearisation has an exact characteristic polynomial; under positive parameters the stationary instability is infrared-first and occupies a finite band of physical wavenumber. A gauge-invariant scalar–Maxwell kinetic completion then supplies a pump for finite- k photons. Reciprocal conformal matching across a crossover surface maps outgoing de Sitter scale-factor asymptotics to an incoming radiation-era law $a \propto t^{1/2}$, while regular conformal Maxwell data appear cold on the outgoing side and hot on the incoming side. We call the resulting picture *Hysterical Conformal Cyclic Cosmology* (HCCC): geometrically akin to Penrose’s conformal cyclic cosmology, but driven by an *active* electromagnetic phase transition rather than a passive conformal identification. That crossover is the big-bang-like event of the next aeon. We separate algebraic theorems from modelling closures and from open dynamical questions (saturation amplitude, junction regularity, thermalisation).

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1 Scope and status

The response-theory paper isolates the Hysterical Response as a residual of an already-existing interaction under exponentially stable reduced dynamics [1]. The late-universe continuation shows that homogeneous dilution of a conserved density, together with a relaxation gap vanishing as a positive power of that density, forces a unique criticality scale and a positive vacuum-like attractor [2]. That same late-universe paper records a sharp electromagnetic caveat: exact charge conservation forbids generating a homogeneous net charge from $Q = 0$, while finite-wavenumber electromagnetic segregation remains allowed.

The present paper develops that electromagnetic window into a cosmological mechanism. In deep late-time de Sitter, an infrared electromagnetic response channel can become supercritical. The resulting phase transition pumps conformal radiation data and, under a reciprocal conformal matching hypothesis of Penrose–Tod type [3, 4, 6], supplies the initial radiation content of a subsequent aeon. The claim is interpretive but sharp:

$$\boxed{\begin{array}{l} \text{IR EM Hysterical criticality} + \text{photon pump} + \text{reciprocal conformal matching} \\ \implies \text{active big-bang-like crossover to a new aeon.} \end{array}} \quad (1)$$

We call this package Hysterical Conformal Cyclic Cosmology (HCCC). It is deliberately compared with Penrose’s conformal cyclic cosmology (CCC) [3, 4, 5]: both identify the remote future of one aeon with the Big Bang of the next through conformal geometry. The distinguishing thesis is dynamical. Standard CCC is *passive*: after massive degrees of freedom become irrelevant, conformal invariance licenses a smooth identification across the crossover [4, 6]. HCCC is *active*: a Hysterical infrared electromagnetic instability drives a phase transition that produces the conformal radiation data of the next aeon.

Four logical layers must be kept separate.

Theorem	Algebraic or elementary-analytic consequence of the stated hypotheses.
Modelling assumption	Scalar–Maxwell kinetic completion; reciprocal conformal ansatz; FLRW asymptotics.
Physical input / literature	Expanding late-time de Sitter; 4D Maxwell conformal invariance; CCC geometry.
Not claimed	Derivation of the saturation amplitude; full Einstein–Maxwell–response junction; entropy production; observational CCC rings/Hawking points; proof that Nature realises HCCC.

2 Inherited late-universe setting

We work on a spatially flat FLRW background with scale factor $a(t)$ and Hubble rate $H = \dot{a}/a > 0$. In the late-universe gravitational channel of [2], dilution forces a unique criticality scale a_c under a density-vanishing gap law, after which a pitchfork attractor yields a positive asymptotic density with $w \rightarrow -1$ from below. That package is inherited here only as background motivation: once a vacuum-like term dominates, the outgoing geometry is asymptotically de Sitter,

$$a_-(\eta) = \frac{1}{H_-|\eta|} + o(|\eta|^{-1}), \quad \eta \rightarrow 0^-, \quad (2)$$

in conformal time η with constant $H_- > 0$. We do not re-prove eventual gravitational criticality.

The electromagnetic distinction of [2] remains in force. Exact charge conservation still forbids a homogeneous net-charge instability from $Q = 0$. The mechanism below is therefore a field / finite-wavenumber electromagnetic response, not creation of cosmic net charge.

3 Linear infrared electromagnetic instability

For one helical Fourier mode with physical wavenumber $q = k/a$ and helicity $s \in \{-1, +1\}$, consider the three-variable linear response model

$$\dot{E} = -2HE + sqB + \alpha h, \quad (3)$$

$$\dot{B} = -2HB - sqE, \quad (4)$$

$$\dot{h} = \sigma E - \gamma h. \quad (5)$$

Here $H > 0$ is treated as locally constant (de Sitter freezing), $\gamma > 0$ is the response relaxation rate, and only the product $A := \alpha\sigma$ enters the spectrum. The Jacobian is

$$J = \begin{pmatrix} -2H & sq & \alpha \\ -sq & -2H & 0 \\ \sigma & 0 & -\gamma \end{pmatrix}. \quad (6)$$

Proposition 3.1 (Characteristic polynomial). *The characteristic polynomial of (6) is independent of helicity and equals*

$$p_q(\lambda) = (\lambda + \gamma)((\lambda + 2H)^2 + q^2) - A(\lambda + 2H). \quad (7)$$

In particular,

$$p_q(0) = \gamma(4H^2 + q^2) - 2HA. \quad (8)$$

Proof. Direct expansion of $\det(\lambda I - J)$; the helicity terms cancel in the characteristic polynomial. Evaluating at $\lambda = 0$ gives (8). $\square$

Theorem 3.2 (Zero-mode stationary threshold). *Assume $H \neq 0$. Then*

$$p_0(0) = 0 \iff A = 2H\gamma. \quad (9)$$

Proof. Set $q = 0$ in (8). $\square$

Define the stationary electromagnetic gain

$$G_{\text{EM}}(q) = \frac{A}{\gamma(2H + q^2/(2H))} \quad (10)$$

and the stationary critical relaxation rate

$$\gamma_c(q) = \frac{2HA}{4H^2 + q^2}. \quad (11)$$

Theorem 3.3 (Infrared-first stationary criticality). *Assume $A, H, \gamma > 0$. Then $G_{\text{EM}}(q)$ is strictly decreasing in q^2 , and*

$$\gamma_c(q^2) \leq \gamma_c(0) = \frac{A}{2H}, \quad (12)$$

with equality only at $q = 0$. Equivalently, among stationary zero-eigenvalue crossings, the $q = 0$ mode reaches criticality first as γ decreases (or A increases).

Proof. Differentiating (10) in q^2 yields a strictly negative quantity for positive parameters. The comparison for γ_c is immediate from (11) since the denominator is minimised at $q = 0$. $\square$

Theorem 3.4 (Infrared unstable band). *Assume $H, \gamma > 0$. Then*

$$p_q(0) < 0 \iff q^2 < \frac{2HA}{\gamma} - 4H^2. \quad (13)$$

In particular, if $A > 2H\gamma$, the set of q with $p_q(0) < 0$ is a nonempty open infrared band $0 \leq q < q_c$ with

$$q_c^2 = \frac{2HA}{\gamma} - 4H^2 > 0. \quad (14)$$

Since $p_q(\lambda) \rightarrow +\infty$ as $\lambda \rightarrow +\infty$, the sign test $p_q(0) < 0$ guarantees a positive real root of p_q .

Proof. Rearrange (8). The limit $p_q(\lambda) \rightarrow +\infty$ as $\lambda \rightarrow +\infty$ is immediate from the cubic leading term, so a negative value at $\lambda = 0$ forces a positive real root by the intermediate-value theorem. $\square$

Remark 3.5 (Stationary versus Hopf). The zero-eigenvalue test does not by itself exclude an earlier Hopf crossing for arbitrary parameter paths. Theorem 3.3 is a stationary-branch statement: along the common experiment in which γ decreases through the zero-mode threshold, infrared modes go unstable first. A global first-instability theorem for all paths requires the cubic Routh–Hurwitz criterion on (7); we do not claim that stronger result here.

4 Near-threshold growth

Write $A = 2H\gamma(1 + \epsilon)$ with $0 < \epsilon \ll 1$. Taylor expansion of $p_q(\lambda) = 0$ about $(\lambda, q^2, \epsilon) = (0, 0, 0)$ yields the leading dispersion

$$\lambda(q) \simeq \frac{\gamma}{2H(2H + \gamma)}(4H^2\epsilon - q^2). \quad (15)$$

Thus the leading unstable band is $q < 2H\sqrt{\epsilon}$, and the maximum growth rate occurs at $q = 0$.

Proposition 4.1 (Infrared onset, algebraic form). *Under the linearised threshold parametrisation $A = 2H\gamma(1 + \epsilon)$ and the frozen denominator approximation of (15), one has $\lambda(q) > 0$ if and only if $q^2 < 4H^2\epsilon$, with maximal growth at $q = 0$.*

Proof. Immediate from the sign of (15) for positive γ, H . $\square$

This is an infrared onset rather than an ultraviolet catastrophe: the linear instability selects the longest available physical wavelengths first.

5 Nonlinear gauge-invariant completion

Modeling Assumption 5.1 (Scalar–Maxwell kinetic completion). A minimal covariant completion uses a scalar response h and a field-dependent Maxwell kinetic function,

$$S = \int d^4x \sqrt{-g} \left[-\frac{1}{2}(\nabla h)^2 - V(h) - \frac{1}{4}I(h)^2 F_{\mu\nu} F^{\mu\nu} \right]. \quad (16)$$

A post-critical Landau potential

$$V(h) = -\frac{1}{2}\mu^2 h^2 + \frac{\lambda_h}{4}h^4, \quad \mu^2, \lambda_h > 0, \quad (17)$$

saturates at $|h_\star| = \mu/\sqrt{\lambda_h}$ in the absence of strong backreaction.

In conformal FLRW coordinates and Coulomb gauge, a transverse Fourier mode with canonical variable $u_k = I A_k$ obeys

$$u_k'' + \left(k^2 - \frac{I''}{I}\right)u_k = 0. \quad (18)$$

Consequently a non-adiabatic homogeneous response is an electromagnetic pump. A sufficient interval-wise condition for tachyonic growth is $I''/I > k^2$. If h oscillates about its minimum, the periodic part of I''/I produces a Hill equation and may generate parametric resonance.

Remark 5.2 (What the pump does not prove). Equation (18) identifies a concrete channel from an infrared response instability to finite- k photon occupation. It does not prove reheating. Efficiency depends on the profile of $I(h)$, expansion, rescattering, strong-coupling constraints, and backreaction. The electromagnetic energy must be computed self-consistently with the h equation and with Einstein dynamics. Those steps are modelling closures and open dynamical problems, not theorems of this paper.

6 Reciprocal conformal matching

Let a regular “bridge” metric $\hat{g}$ describe a neighbourhood of a spacelike crossover Σ , and write

$$g_{\mu\nu}^- = \Omega_-^2 \hat{g}_{\mu\nu}, \quad g_{\mu\nu}^+ = \Omega_+^2 \hat{g}_{\mu\nu}. \quad (19)$$

Assume $\Omega_- \rightarrow \infty$ and $\Omega_+ \rightarrow 0$ as the crossover is approached from the outgoing and incoming sides respectively, and impose the reciprocal ansatz

$$\Omega_+ \Omega_- = C_0, \quad C_0 > 0. \quad (20)$$

This is the geometric skeleton shared with Penrose CCC [3, 4, 6]: future conformal infinity of one aeon is identified with the Big Bang of the next.

Assumption 6.1 (Four-dimensional Maxwell conformal transport). In four spacetime dimensions, source-free Maxwell equations are conformally invariant when the two-form $F_{\mu\nu}$ is held fixed [7]. The covariant stress tensor and measured density then scale as

$$T_{\mu\nu} = \Omega^{-2} \hat{T}_{\mu\nu}, \quad \rho = \Omega^{-4} \hat{\rho}. \quad (21)$$

Theorem 6.2 (Cold outgoing / hot incoming). *Under Assumption 6.1 and reciprocity (20), finite nonzero conformal radiation density $\hat{\rho} > 0$ satisfies*

$$\rho_- = \Omega_-^{-4} \hat{\rho} \rightarrow 0, \quad \rho_+ = \Omega_+^{-4} \hat{\rho} \rightarrow \infty \quad (22)$$

as $\Omega_- \rightarrow \infty$ and $\Omega_+ \rightarrow 0$ with $\Omega_+ \Omega_- = C_0$. Equivalently, with unit reciprocity $a_+ a_- = 1$ and fixed $\hat{\rho} \neq 0$,

$$\frac{\rho_+}{\rho_-} = a_-^8. \quad (23)$$

Proof. The scalings (21) give $\rho_{\pm} = \Omega_{\pm}^{-4} \hat{\rho}$. Reciprocity forces $\Omega_+ = C_0/\Omega_-$, hence $\rho_+/\rho_- = (\Omega_-/\Omega_+)^4 = \Omega_-^8/C_0^4$. In FLRW units with $a_{\pm} \propto \Omega_{\pm}$ and $C_0 = 1$, this is (23). $\square$

Remark 6.3 (Interpretation). Theorem 6.2 is a relation between two physical metrics and common conformal data. It must not be read as ordinary energy amplification inside a single metric. On the outgoing side the same conformal radiation looks arbitrarily cold; on the incoming side it looks arbitrarily hot. That hot incoming branch is the big-bang-like content of the next aeon.

Theorem 6.4 (Reciprocal de Sitter $\rightarrow$ radiation asymptotics). *Assume the outgoing asymptotic (2) and unit reciprocity $a_+(\eta) = 1/a_-(-\eta)$ for $\eta > 0$ small. Then*

$$a_+(\eta) = H_- \eta + o(\eta). \quad (24)$$

Writing $dt_+ = a_+ d\eta$ and $a_+ = C\eta$ with $C = H_-$, one obtains

$$a_+^2 = 2C t_+, \quad (25)$$

hence the radiation-era FLRW exponent $a_+ \propto t_+^{1/2}$.

Proof. Substitute (2) into the reciprocal ansatz. With $a = C\eta$ and $t = \int a d\eta = C\eta^2/2$ one has $a^2 = 2Ct$ by direct algebra. $\square$

Proposition 6.5 (Reciprocal physical momenta). *If $a_+ a_- = 1$ and $a_- \neq 0$, then $k/a_+ = k a_-$ for any comoving wavenumber k . Soft outgoing physical momenta are therefore hard on the incoming side.*

Proof. Immediate from $1/a_+ = a_-$. $\square$

Remark 6.6 (Normalization caveat). Reciprocity alone fixes an asymptotic form only up to normalization. If one additionally assumes a flat radiation-dominated Friedmann equation and writes $a_+ = C\eta$, then $\hat{\rho}_\gamma = 3C^2/(8\pi G)$. Setting $C = H_-$ by unit reciprocity gives $\hat{\rho}_\gamma = 3H_-^2/(8\pi G)$. That equality is a consistency condition involving conformal gauge and Einstein dynamics, not an independent prediction of Maxwell conformal invariance alone.

7 Active versus passive conformal cyclic cosmology

Penrose's CCC proposes an infinite succession of aeons in which the conformal infinity of one aeon is identified with the Big Bang of the next [3, 4, 5]. The geometric matching—a spacelike crossover, conformal factors diverging/vanishing, and transport of zero-rest-mass field data—is the shared spine of CCC and HCCC.

The dynamical reading differs.

	Penrose CCC	HCCC (this paper)
Geometry	Conformal aeon bridge	Same reciprocal conformal bridge
Driver	Passive: massless conformal era after matter evaporates	Active: IR EM Hysterical phase transition
Radiation seed	Inherited conformal data / Hawking evaporation products	Photon pump from supercritical response (18)
Big Bang of next aeon	Conformally reinterpreted future infinity	Hot incoming branch of Theorem 6.2
Status claimed here	Literature geometry	Geometry + IR instability theorems; pump/junction open

Definition 7.1 (Active big-bang-like crossover). A crossover Σ is *active big-bang-like* when:

1. an outgoing aeon approaches a de Sitter-like conformal infinity;
2. an instability produces nonzero conformal radiation data $\hat{\rho}$ on a neighbourhood of Σ ;
3. reciprocal conformal matching maps those data to an incoming radiation-dominated asymptotics with $\rho_+ \rightarrow \infty$ as in Theorem 6.2.

Under Modelling Assumption 5.1, Assumption 6.1, and the reciprocal ansatz, the infrared electromagnetic mechanism of Sections 3–6 supplies a concrete candidate for Definition 7.1. The phase transition is the physical event; the conformal matching is the geometric translation of that event into the language of the next aeon’s Big Bang.

Remark 7.2 (What is and is not identified with “our” Big Bang). HCCC claims that *some* aeon-to-aeon crossover of this type is big-bang-like for the incoming side. Whether the Big Bang of our observed aeon was produced by a previous Hysterical electromagnetic crossover is a physical hypothesis, not a theorem. Likewise, observational CCC proposals (concentric low-variance circles, Hawking points) are not evaluated here [4, 5].

8 Logical status of the results

Quantity / claim	Status
Characteristic polynomial (7) from $\det(\lambda I - J)$	Prop. 3.1 (Lean: <code>prop_char</code> , helicity $s^2 = 1$).
Zero-mode threshold $A = 2H\gamma$	Theorem 3.2 (Lean).
IR-first stationary criticality (incl. strict)	Theorem 3.3 (Lean: <code>thm_IR_first_full</code>).
Infrared unstable band / sign test / positive root	Theorem 3.4 (Lean: <code>thm_band_full</code> , IVT).
Near-threshold dispersion sign	Prop. 4.1 (Lean algebra of the frozen-denominator form).
Scalar–Maxwell action / $I(h)$ pump	Modelling Assumption 5.1; not Lean.
Photon production efficiency / reheating	Open dynamical problem.
4D Maxwell conformal invariance	Literature hypothesis [7].
Cold/hot density dichotomy; $\rho_+/\rho_- = a^8_-$	Theorem 6.2 (Lean: unit and C_0 forms).
Reciprocal dS→radiation (exact leading / linear ansatz)	Theorem 6.4 (Lean: <code>thm_radiation_full</code> ; little- o remainder not re-proved).
Reciprocal momenta	Prop. 6.5 (Lean).
Reciprocal conformal ansatz $\Omega_+\Omega_- = C_0$	Modelling / CCC geometry input.
Saturation value $\hat{\rho}_{\text{EM}}(\Sigma)$	Not derived.
Einstein–Maxwell–response junction regularity	Open.
Entropy production / thermalisation	Open.
Observational CCC signatures	Not claimed.
“Nature realises HCCC”	Not claimed.

9 Conclusion

The late-universe gravitational package of this series leaves an electromagnetic window: finite-wavenumber response can become critical even though homogeneous net charge cannot be created from neutrality. In an asymptotically de Sitter background, a minimal helical electromagnetic–response model has an infrared-first stationary instability and a finite unstable band. A scalar-dependent Maxwell kinetic function then offers a gauge-invariant route from that infrared onset to finite- k photons. Reciprocal conformal matching—the geometric core shared with Penrose’s conformal cyclic cosmology—maps outgoing de Sitter asymptotics to an incoming radiation-era law and converts regular conformal radiation data into a hot big-bang-like branch for the next aeon.

The resulting Hysterical Conformal Cyclic Cosmology is therefore CCC-like in conformal geometry and distinct in dynamics: the crossover is an active phase transition, not a passive conformal identification after the last massive degrees of freedom fade. The theorems close the algebraic spine of that claim. The scientific burden that remains is dynamical: derive the saturated conformal radiation amplitude, solve the coupled response–Maxwell–Einstein system through the junction, and

demonstrate approximate thermalisation. Until those steps are closed, HCCC is a mathematically structured mechanism for an active aeon bridge, not a completed cyclic cosmology of Nature.

References

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